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Seasonality and approximation errors in rational expectations models*

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A frequency domain representation of the approximation criterion that is implicit in Gaussian maximum likelihood estimation is applied to study the effects of using seasonally adjusted versus seasonally unadjusted data to estimate rational expectations models. Three classes of economic mechanisms for generating seasonality are described. Approximating parameter estimates are computed numerically for several examples.

1. Introduction

This paper illustrates a framework for evaluating errors in fitting a dynamic equilibrium model via maximum likelihood when it is assumed that the model is an approximation. By extending an approach taken by Sims (1972), we derive a frequency domain expression for the criterion that maximum likelihood estimates of an approximating model implicitly minimize. The parameter values that minimize this approximation criterion are the probability limits of misspecified maximum likelihood estimators.

This paper uses the approximation criterion to compare the merits of seasonally adjusted and seasonally unadjusted data for estimating an equilibrium (or rational expectations) model. To this end, the approximation criterion is designed to focus on alternative ways of modelling seasonality: both the true and the approximating models are permitted, though not required, to have periodic means and autocovariances for observables. We use our apparatus to evaluate an argument of Sims (1976) that rational expectations models should be estimated with seasonally adjusted data. Other writers[1] had

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[1]See Sargent (1978) and Singleton (1988).

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asserted that seasonally unadjusted data ought to be used to estimate rational expectations models, on the grounds that the agents being modelled were themselves assumed to be solving forecasting and signal extraction problems using seasonally *unadjusted* data, and that the cross-equation restrictions implied by their behavior applied to seasonally unadjusted series. Sims responded that that argument assumes that the model is correctly specified, and asserted that admitting the possibility of approximation error would alter the situation. Sims said that if approximation error were particularly bad with respect to the seasonal components of a model, better estimates of at least a subset of economically relevant parameters might be attained by using seasonally adjusted data and giving up the pretense of accurately modelling all frequencies.

Section 2 briefly describes the class of stochastic models that we consider, a class which includes the periodic models that have been studied in various respects by Gladysev (1960) and Tiao and Grupe (1980). Section 3 describes a class of *economic* models whose equilibria assume the form of the statistical model of section 2. These economic models are rich enough to illustrate three possible models of seasonality that have been described by Hansen and Sargent (1992). Section 4 describes the approximation criterion that is minimized by maximum likelihood estimates of a time-invariant model (possibly allowing seasonal means) when the data are truly generated by a particular alternative model. Section 5 applies the criterion to several examples. Conclusions and extensions are discussed in section 6.

The particular examples in section 5 seem to bolster a recommendation for using seasonally *adjusted* data from two points of view. First, some of our examples show that when the mechanism generating seasonality is misspecified, superior estimates of economic parameters are made with seasonally adjusted data. These examples are somewhat special because they are structured in such a way that the true model and the approximating model share some common parameters, better estimates of which are obtained with seasonally adjusted data. (Obviously, it is easy to create examples in which the approximating and the true models share no common parameters, making it more problematical to interpret how these parameters compare.)[2]

Second, some of our examples illustrate that in situations where the model is correctly specified, using seasonally adjusted data induces no asymptotic biases even when some of the key parameters are those governing behavior primarily at the seasonal frequencies. Strong cross-equation, cross-frequency overidentifying restrictions permit identification of *all* parameters, even when seasonal frequencies are deemphasized (or even deleted) via seasonal adjustment.

[2]Examples that illustrate a prejudice in favor of using seasonally unadjusted data are given in Sargent (1987, problems 22, 23, pp. 355–357).

The examples in section 5 are enough to induce skepticism against any *general* presumption in favor of using seasonally unadjusted data. However, the examples are insufficient to establish a general presumption in favor of always using seasonally adjusted data. Although we supply none in this paper, we do not doubt that examples can be found in which superior estimates of parameters of misspecified models can be obtained using seasonally unadjusted data. The approximation criterion can be put to use to analyze such examples. Also, the result that consistent estimates of correctly specified models will be obtained with seasonally adjusted data has to be interpreted carefully: that the model is correctly specified means that it correctly reproduces *all* frequencies, including the seasonal frequencies that are deemphasized (or omitted) via seasonal adjustment. Thus, this finding will not by itself rationalize the standard practice of formulating and estimating a model that implies that the data have little seasonal fluctuations.

2. *Covariance-stationary models of seasonality*

This section briefly describes the class of statistical models that we use. The following section describes an illustrative class of *economic* models whose equilibria take this statistical form. We make use of a scalar sequence of 'seasonal indicators'. In particular, let $\{s(t)\}$ denote a *periodic* sequence mapping calendar time t into the season $s(t)$. The periodic character of this sequence is reflected in the restriction $s(t+p)=s(t)$ for all t. We could initialize the sequence by setting $s(t)=t$ for $t=1,2,\dots,p$. Coupled with the periodic structure, this initialization would be sufficient to define the entire sequence $\{s(t)\}$.

However, we shall construct an $\{s(t)\}$ process in a somewhat different way, so that it retains the interpretation of mapping 'time' into 'season', but does so in such a way that it can be viewed as a stationary stochastic process. We accomplish this by *randomizing* the assignment of a season. In particular, we consider p different assignments, each occurring with probability $1/p$. The first assignment is the one just described, and results in a sequence that we now denote $\{s(t,1)\}$. The second assignment is obtained from the first by shifting it forward: we construct $s(t,2)=s(t+1,1)$ and, more generally, $s(t,j+1)=s(t+1,j)$ for $j=1,\dots,p-1$. Then we have p alternative periodic sequences, which differ only in how they match up with the calendar dates. Now imagine an initial lottery in which one of these sequences is drawn at random, so that with probability $1/p$, sequence $\{s(t,j)\}$ is drawn. With this initial lottery, the resulting process, which we call $\{s(t)\}$, is strictly stationary and ergodic.

Some of our calculations will be conducted by conditioning on the season. In spite of the randomization, the process is *deterministic*, meaning that the process is perfectly predictable given its present value. Hence conditioning on the entire process s is equivalent to conditioning on the process at some date, say $s(t)$. Abusing notation slightly, we let either $\mathrm{E}(\cdot\,|\,s)$ or $\mathrm{E}[\cdot\,|\,s(t)]$ denote the expectations operator based on such conditioning.

We now use the stochastic process $\{s(t)\}$ together with an $(n \times 1)$ stationary, ergodic martingale difference sequence $\{w_t\}$ conditioned on s to build a class of *periodic*, linear time series models. We assume that the martingale difference sequence is conditionally homoskedastic, i.e., $\mathrm{E}(w_t w_t' | w_{t-1}, w_{t-2}, \ldots; s) = I$, and we consider models of the form

$$y_t = M_{s(t)}(L) w_t + \nu_{s(t)}, \tag{2.1}$$

where $\{y_t\}$ is an $(m \times 1)$ stochastic process of variables observable to an econometrician; L is the lag operator; $M_{s(t)}(L) = \sum_{j=0}^{\infty} M_{s(t),j} L^j$, $\sum_{j=0}^{\infty} \operatorname{trace} M_{s(t),j} M_{s(t),j}' < +\infty$; $\nu_{s(t)}$ is an $(m \times 1)$ vector of (seasonal) means. Note that the moving average representation for the process $\{y_t\}$ is specified to depend on the seasonal process $\{s(t)\}$. Since $\{s(t)\}$ and $\{w_t\}$ are (jointly) stationary and ergodic, so is $\{y_t\}$. This is convenient because it is known that such processes obey a Law of Large Numbers [e.g., see Breiman (1968)].[3]

For a realization of data, $\{y_t\}_{t=1}^{T}$, consider two distinct ways of forming sample moments. We can compute sample moments either conditioned on the season, say $s(t) = j$, or unconditionally. Sample moments conditioned on the season are computed by *skip-sampling* with a skip interval of length p. So long as the population moments exist and are finite, the Law of Large Numbers for stationary processes implies that the sample moments converge as the sample size gets large. The limit points all have interpretations as *population expectations*, either conditional or unconditional. The limit points of the skip-sampled averages may differ depending on the season, and the limit points of the unconditional sample moments will just be the simple average of those seasonal limit points. We shall focus on means and covariances.

Consider first the means. For model (2.1) it can be established that

$$\lim_{T\to\infty} \frac{1}{T} \sum_{\tau=1}^{T} y_{t+\tau p} = \mathrm{E}[\,y_t | s(t)] = \nu_{s(t)},$$

where the limit on the left-hand side is an almost sure limit. Notice that in forming the above means, we average only over y_t's in the same season of the

[3]Sometimes, periodic models have been referred to as embodying a particular kind of *nonstationarity*; for example, see Tiao and Grupe (1980). Our randomization over the initial season eliminates nonstationarity while leaving intact the periodicity. In Hansen and Sargent (1992, ch. 12), we provide a formal apparatus for characterizing or *detecting* the presence of periodicity in an arbitrary stationary stochastic process. In particular, we provide a definition of 'periodicity' that is compatible with the present construction of $\{y_t\}$ as a process with periodicity p.

year (i.e., over observations separated by p periods). If we average unconditionally, we obtain

$$\lim_{T\to\infty}\frac{1}{T}\sum_{\tau=1}^{T} y_\tau = \mathrm{E}y_t = \frac{1}{p}\sum_{t=1}^{p}\nu_{s(t)}.$$

Now think of forming sample covariances of observations from the mean adjusted process $\tilde{y}_t = y_t - \nu_{s(t)} = M_{s(t)}(L)w_t$, conditionally and unconditionally. It can be shown that the following limits exist and that they satisfy:

$$\lim_{T\to\infty}\frac{1}{T}\sum_{\tau=1}^{T}\tilde{y}_{t+\tau p}\tilde{y}'_{t+\tau p+k} = \mathrm{E}\left[\tilde{y}_t\tilde{y}'_{t+k}|s(t)\right] = c_{s(t)}(-k), \tag{2.2}$$

$$\lim_{T\to\infty}\frac{1}{T}\sum_{\tau=1}^{T}\tilde{y}_\tau\tilde{y}'_{\tau+k} = \mathrm{E}\tilde{y}_t\tilde{y}'_{t+k} = r(-k) = \frac{1}{p}\sum_{t=1}^{p}c_{s(t)}(-k). \tag{2.3}$$

We briefly indicate how results of Gladysev (1960) and Tiao and Grupe (1980) can be used to deduce the covariance generating function of $\{r(-k)\}$ as a function of the parameters of (2.1). Notice that (2.1) implies

$$\begin{bmatrix} y_{p\cdot t-p+1} \\ y_{p\cdot t-p+2} \\ \vdots \\ y_{p\cdot t-1} \\ y_{p\cdot t} \end{bmatrix}$$

$$= \begin{bmatrix} \sum_{j=0}^{\infty} M_{1,p\cdot j}L^{p\cdot j} & \sum_{j=0}^{\infty} M_{1,p\cdot j+p-1}L^{p\cdot(j+1)} & \cdots & \sum_{j=0}^{\infty} M_{1,p\cdot j+2}L^{p\cdot(j+1)} & \sum_{j=0}^{\infty} M_{1,p\cdot j+1}L^{p\cdot(j+1)} \\ \sum_{j=0}^{\infty} M_{2,p\cdot j+1}L^{p\cdot j} & \sum_{j=0}^{\infty} M_{2,p\cdot j}L^{p\cdot j} & \cdots & \sum_{j=0}^{\infty} M_{2,p\cdot j+3}L^{p\cdot(j+1)} & \sum_{j=0}^{\infty} M_{2,p\cdot j+2}L^{p\cdot(j+1)} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \sum_{j=0}^{\infty} M_{p-1,j+p-2}L^{p\cdot j} & \sum_{j=0}^{\infty} M_{p-1,p\cdot j+p-3}L^{p\cdot j} & \cdots & \sum_{j=0}^{\infty} M_{p-1,p\cdot j}L^{p\cdot j} & \sum_{j=0}^{\infty} M_{p-1,p\cdot j+p-1}L^{p\cdot(j+1)} \\ \sum_{j=0}^{\infty} M_{p,p\cdot j+p-1}L^{p\cdot j} & \sum_{j=0}^{\infty} M_{p,p\cdot j+p-2}L^{p\cdot j} & \cdots & \sum_{j=0}^{\infty} M_{p,p\cdot j+1}L^{p\cdot j} & \sum_{j=0}^{\infty} M_{p,p\cdot j}L^{p\cdot j} \end{bmatrix}$$

$$\begin{bmatrix} w_{p\cdot t-p+1} \\ w_{p\cdot t-p+2} \\ \vdots \\ w_{p\cdot t-1} \\ w_{p\cdot t} \end{bmatrix} + \begin{bmatrix} \nu_1 \\ \nu_2 \\ \vdots \\ \nu_{p-1} \\ \nu_p \end{bmatrix}.$$

We write the above equation as

$$Y_t = \bar{M}(L)W_t + \nu, \qquad t = 1,2,\ldots \tag{2.4}$$

where

$$\bar{M}(L) = \sum_{j=0}^{\infty} \bar{M}_j L^j,\ \sum_{j=0}^{\infty} \text{trace}\ \bar{M}_j \bar{M}_j' < +\infty,$$

$$Y_t' = [y_{p\cdot t-p+1}', y_{p\cdot t-p+2}', \ldots, y_{p\cdot t}'],$$

$$W_t = [w_{p\cdot t-p+1}', w_{p\cdot t-p+2}', \ldots, w_{p\cdot t}']',$$

$$\nu' = [\nu_1', \nu_2', \ldots, \nu_p'].$$

Evidently, W_t is a martingale difference sequence (adapted to the appropriate sampled sequence of information sets) with a contemporaneous covariance matrix equal to an identity matrix. By construction, the stacked process $\{Y_t\}$ has periodicity equal to one. The covariance generating function of the process $(Y_t - \nu)$ is

$$S_Y(z) = \bar{M}(z)\bar{M}(z^{-1})'. \tag{2.5}$$

Let $s_y(z) = \sum_{k=-\infty}^{\infty} z^{-k} r(-k)$. Tiao and Grupe (1980) showed that

$$s_y(z) = Q(z^{-1})' S_Y(z^p) Q(z), \tag{2.6}$$

where

$$Q(z) = \frac{1}{\sqrt{p}}\left[Iz^p \vdots Iz^{p-1} \vdots \cdots \vdots Iz\right]'.$$

Formula (2.6) is useful because it characterizes the restrictions that a periodic model imposes on second moments that are constructed in the ordinary fashion, by not conditioning on season of the year. Formula (2.6) will play a role in our application of the criterion that we develop in section 4 to characterize the misinterpretations that a model builder will make if he mistakenly ignores hidden periodicity when it is truly present.

3. Economic models of seasonality

In this section, we describe an economic model whose equilibrium assumes the mathematical form of eq. (2.1). We use the model to illustrate three alternative ways of economically interpreting seasonality in prices and quantities. We formulate the model as a planning model, though it also has a competitive equilibrium interpretation.[4]

[4]See Stokey, Lucas, and Prescott (1989) or Hansen and Sargent (1992) for the connection between the planning problem and a competitive equilibrium.

A fictitious social planner chooses contingency plans $\{c_t, k_t, i_t\}_{t=0}^{\infty}$ to maximize the utility functional

$$-\tfrac{1}{2}\mathrm{E}\sum_{t=0}^{\infty}\beta^t\left[\left(c_t-\lambda(1-\delta_h)\sum_{j=1}^{\infty}\delta_h^{j-1}c_{t-p\cdot j}-b_t\right)^2+l_t^2\right]|I_0, \tag{3.1}$$

$$0\le\lambda\le 1,\quad 0<\delta_h<1,\quad 0<\beta<1,$$

subject to the technology

$$c_t+i_t=\gamma_{s(t)}k_{t-1}+d_t,\qquad \gamma_{s(t)}\ge 0,$$

$$k_t=\delta_k k_{t-1}+i_t,\qquad 0<\delta_k<1,$$

$$\phi_1 i_t=|l_t|,\qquad \phi_1>0,$$

$$s(t+p)=s(t),\quad \forall t,\qquad s(t)=t\quad\text{for}\quad t=1,\dots,p, \tag{3.2}$$

and subject to the (exogenous) laws of motion

$$a_b(L)b_t=w_{1t}+\mu_b,\qquad a_d(L)d_t=w_{2t}+\mu_d, \tag{3.3}$$

where $a_b(L)=\sum_{j=0}^{n}a_{bj}L^j$, $a_d(L)=\sum_{j=0}^{n}a_{dj}L^j$, and where $a_b(z_0)=0\Rightarrow|z_0|>1$ and $a_d(z_0)=0\Rightarrow|z_0|>1$. In (3.3), $w_t=[w_{1t}\ \ w_{2t}]'$ is a martingale difference sequence with $\mathrm{E}[w_t w_t'|I_{t-1};s]=I$, where I_t is the information set that the planner has at time t, which includes the history of consumption and capital through time $t-1$ and the histories of the processes $\{b_t, d_t\}$ through time t. Here k_t is the capital stock, l_t is labor supply, and i_t is the investment rate. There are two random disturbances impinging on the model, both of which are determined by univariate autoregressive processes: $\{b_t\}$ is a stochastic process of shocks to preferences and $\{d_t\}$ is a stochastic process of endowment disturbances.

This is a linear quadratic version of a stochastic one-sector optimum growth model of a kind introduced by Brock and Mirman (1972), with preferences modified somewhat in the manner of Ryder and Heal (1973).[5] We have added labor costs of adjusting the capital stock, for technical reasons described by Hansen and Sargent (1992, ch. 3).[6] Let $y_t=[c_t\ \ i_t]'$ be a vector of variables from this economy observable to an econometrician. Then the solution of the social planning problem takes the form of our eq. (2.1).

[5]Model (3.1)–(3.3) is a special case of a class of linear quadratic models described by Hansen and Sargent (1992).

[6]This makes the model interpretable as a general equilibrium version of Lucas and Prescott's (1971) model.

When the productivity parameter $\gamma_{s(t)}$ is time-invariant ($\gamma_{s(t)} = \gamma$, $\forall t$), the moving average kernel $M_{s(t)}(L)$ is time-invariant (independent of t) and so are the means $\nu_{s(t)}$.

The structure (3.1)–(3.3) is sufficiently rich that it can illustrate three alternative kinds of models of the sources of seasonality. We briefly describe each of these.

(a) *Time-invariant models in which the seasonality stems from seasonality in exogenous shock processes.* Set $\lambda = 0$ and $\gamma_{s(t)} = \gamma$, $\forall t$, so that the 'seasonal habit persistence' term $(1-\delta_h)\sum_{j=1}^{\infty}\delta_h^{j-1}c_{t-p\cdot j}$ is excluded from the utility functional and the productivity parameter is time-invariant. However, let $a_b(L)$ or $a_d(L)$ be specified so that $\{b_t\}$ or $\{d_t\}$ is a process with spectral peaks at seasonal frequencies. The seasonality in $\{b_t\}$ or $\{d_t\}$ will be transmitted to $y_t = [c_t \;\; i_t]'$ in ways determined by the solution of the social planning problem.

(b) *Time-invariant models in which seasonality comes from a 'propagation mechanism' determined by preferences or technology.* Continue to assume that $\gamma_{s(t)} = \gamma$, $\forall t$, but now activate the seasonal habit persistence term by setting $1 \geq \lambda > 0$, $1 > \delta_h > 0$. Let $a_b(L)$ and $a_d(L)$ both be such that neither $\{b_t\}$ nor $\{d_t\}$ displays spectral peaks at seasonal frequencies. The existence of seasonal habit persistence can induce the planner to put spectral peaks in $\{y_t\}$ at the seasonal frequencies.[7]

(c) *Periodically time-varying models in which the source of seasonality is seasonal periodicity in a technology or preference parameter.* We can illustrate this type of model by setting $\lambda = 0$ and setting $a_b(L)$ and $a_d(L)$ so that neither $\{b_t\}$ nor $\{d_t\}$ has spectral peaks at seasonal frequencies. However, now let $\gamma_{s(t)}$ vary periodically. This implies a solution of the social planning problem of the form (2.1), which via the Tiao–Grupe formula (2.6) can transmit seasonality to $\{y_t\}$.[8]

In section 5 we shall use examples of each of these three types of model to illustrate some consequences of various types of approximation error. In the process, we shall illustrate the way that each kind of model can lead to

[7]This specification is a modification of one that was introduced by Ryder and Heal (1973) and utilized by Becker and Murphy (1988). Heaton (1990) fit models with seasonal habit persistence to seasonally unadjusted data.

[8]Periodic models have been used by Osborn (1988, 1990), Osborn and Smith (1989), Osborn, Chui, Smith, and Birchenhall (1988), and Birchenhall, Bladen-Hovell, Chui, Osborn, and Smith (1989). Todd (1983, 1991) studied rational expectations models whose equilibria assume the form of the periodic linear quadratic models described in section 2. Statistical analysis of periodic models is contained in papers by Jones and Brelsford (1967), Pagano (1978), Monin (1963), Cleveland and Tiao (1979), and Burridge and Wallis (1990).

'spectral peaks at seasonal frequencies in the spectral densities of observables.

4. The approximation criterion

We want to study the errors of economic interpretation that an econometrician makes when he estimates a misspecified model. It is well-known that there is an important link between likelihood estimation and the Kullback–Leibler (1951) information criterion. For instance, Akaike (1973) and White (1982) interpreted and justified the use of the misspecified maximum likelihood estimator as an estimator of the parameter vector which minimizes this information criterion. White's analysis assumed that the data vector is independent and identically distributed but allowed for great flexibility in both the parameterized and true distribution functions. Of more direct relevance for us are the works of Ljung (1979), Ljung and Caines (1979), and Pötscher (1987) because our focus is on stationary linear time series models estimated using a Gaussian likelihood function. We represent a counterpart to the information criterion in terms of a spectral decomposition. In the following section, we use this criterion to study approximation error in the context of some models described in section 3.

We use the stacked process $\{Y_t\}$ described in section 2 in order to eliminate the periodic probability structure. As in section 2, ν is the population mean of this process and $F(\omega) = S_Y[\exp(-i\omega)]$ is the spectral density function at frequency ω. The econometrician fits an approximating model whose free parameters are listed in the vector δ. Let $\mu(\delta)$ denote the mean of this approximating model and $G(\cdot, \delta)$ the spectral density function. According to the approximating model,

$$\mathrm{E}\{[Y_t - \mu(\delta)][Y_{t+\tau} - \mu(\delta)]'\} = \frac{1}{2\pi}\int_{-\pi}^{\pi} \exp(-i\omega\tau)G(\omega,\delta)\,\mathrm{d}\omega. \tag{4.1}$$

The econometrician is assumed to estimate δ by maximum likelihood using a Gaussian likelihood function for $\{Y_t\}$. In the appendix we use results from Hannan (1973), Dunsmuir and Hannan (1976), Deistler, Dunsmuir, and Hannan (1978), and Pötscher (1987) to establish that the maximum likelihood estimator for the misspecified model converges almost surely to the minimizer of the following approximation criterion:

$$A(\delta) = A_1(\delta) + A_2(\delta) + A_3(\delta), \tag{4.2}$$

where

$$A_1(\delta) \equiv \frac{1}{2\pi}\int_{-\pi}^{\pi} \log\det G(\omega,\delta)\,\mathrm{d}\omega,$$

$$A_2(\delta) \equiv \frac{1}{2\pi}\int_{-\pi}^{\pi} \operatorname{trace}\left[G(\omega,\delta)^{-1}F(\omega)\right]\mathrm{d}\omega,$$

$$A_3(\delta) \equiv [\nu-\mu(\delta)]'G(0,\delta)^{-1}[\nu-\mu(\delta)]. \tag{4.3}$$

The term $A_1(\delta)$ is equal to the logarithm of the determinant of the covariance matrix of the one-step forecast errors in $\{Y_t\}$ as implied by the approximating model for parameter δ. The term $A_2(\delta)$ is the limiting value of a term in the likelihood function constructed as a quadratic form using a vector $Y_t-\nu$ and the inverse of a covariance matrix built from the covariances implied by the approximating spectral density $G(\cdot,\delta)$. The term $A_3(\delta)$ is the limiting value of a quadratic form constructed from the vector of approximation errors in the mean $\nu-\mu(\delta)$ and once again the inverse of the covariance matrix formed from the approximating spectral density $G(\cdot,\delta)$.[9]

In implementing formula (4.3), it is convenient to replace the integrals defining $A_1(\delta)$ and $A_2(\delta)$ by finite Riemann sums.[10] (Of course, other numerical methods for approximating integrals could also be employed.) In particular, we let N be even and use the approximation

$$\hat{A}(\delta) = \frac{1}{N}\sum_{j=0}^{N-1}\left\{\log\det G(\omega_j,\delta) + \operatorname{trace}\left[G(\omega_j,\delta)^{-1}F(\omega_j)\right]\right\}$$
$$+[\nu-\mu(\delta)]'G(0,\delta)^{-1}[\nu-\mu(\delta)], \tag{4.4}$$

where $\omega_j = 2\pi j/N$, $j=0,1,\ldots,N-1$. In computing the sum in (4.4), we can exploit conjugate symmetry of the spectral density matrices to reduce the number of terms in each sum to $(N/2+1)$.

[9]Cochrane (1988) and Sims (1990) derived an alternative representation of the approximation criterion applicable when the innovation covariance matrix and the mean vector are not restricted. Periodicity of the form described in section 2 plays no role in their analyses so that the approximation criterion is applied directly to parameterizations of the process $\{y_t\}$. Their formulas are better designed to display the link between the approximation criterion for autoregressive models and the approximation criterion for time series regression models proposed by Sims (1972).

[10]The introduction of small numerical errors in evaluating A might lead to large errors in approximating the minimizer of A. One way to avoid this problem is to require the Riemann approximation to work uniformly well over the parameter space. Sufficient conditions for this uniformity are given in the appendix (see Assumptions A.1–A.3).

For some of the calculations we are interested in, the approximating model ignores the periodic structure in the conditional (on the season) covariances of $\{y_t\}$. In these circumstances, there are alternative ways to represent the components of A. Let $g(\cdot, \delta)$ denote the approximating spectral density for $\{y_t\}$. Recall that $\{Y_t\}$ is formed by *stacking* the process $\{y_t\}$ and then *sampling* every pth observation. The formal construction of G from g reflects both stacking and sampling:

$$\begin{aligned} &G(p\omega, \delta) \\ &\quad = \sum_{j=0}^{p-1} \alpha(\omega + 2\pi j/p) g[(\omega + 2\pi j/p), \delta] \alpha(-\omega - 2\pi j/p)', \end{aligned} \tag{4.5}$$

where

$$\alpha(\omega) \equiv Q[\exp(-i\omega)], \tag{4.6}$$

where Q was defined when we set forth the Tiao–Grupe formula (2.6). *Stacking* is captured by α in (4.6) and *sampling* by the *folding* in (4.5). Since the columns of each matrix $\alpha(\omega + 2\pi j/p)$ for $j = 0, 1, \ldots, p-1$ are orthonormal, the determinant and inverse of the matrix $G(p\omega, \delta)$ are

$$\log\det G(p\omega, \delta) = \sum_{j=0}^{p-1} \log\det g(\omega + 2\pi j/p, \delta), \tag{4.7}$$

and

$$\begin{aligned} &G(p\omega, \delta)^{-1} \\ &\quad = \sum_{j=0}^{p-1} \alpha(\omega + 2\pi j/p) g(\omega + 2\pi j/p, \delta)^{-1} \alpha(-\omega - 2\pi j/p)'. \end{aligned} \tag{4.8}$$

Substituting (4.7) into expression (4.3) for A_1 gives

$$\begin{aligned} A_1(\delta) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} \log\det G(\omega, \delta)\, \mathrm{d}\omega \\ &= \frac{p}{2\pi} \int_{-\pi}^{\pi} \log\det g(\omega, \delta)\, \mathrm{d}\omega, \end{aligned} \tag{4.9}$$

and substituting (4.8) into expression (4.3) for A_2 gives

$$A_2(\delta) = \frac{1}{2\pi}\int_{-\pi}^{\pi} \operatorname{trace}\left[G(\omega,\delta)^{-1}F(\omega)\right]\mathrm{d}\omega$$

$$= \frac{p}{2\pi}\int_{-\pi}^{\pi} \operatorname{trace}\left[g(\omega,\delta)^{-1}\alpha(-\omega)'F(\omega/p)\alpha(\omega)\right]\mathrm{d}\omega$$

$$= \frac{p}{2\pi}\int_{-\pi}^{\pi} \operatorname{trace}\left[g(\omega,\delta)^{-1}f(\omega)\right]\mathrm{d}\omega, \tag{4.10}$$

where $f(\omega)=s_y[\exp(-i\omega)]$ is the spectral density of $\{y_t\}$ at frequency ω formed after seasonal means have been subtracted. The last equality in (4.10) follows from the Tiao–Grupe formula (2.6). Not surprisingly, the expressions for $A_1(\delta)$ and $A_2(\delta)$ given in (4.9) and (4.10) are just the counterparts of the expressions for $A_1(\delta)$ and $A_2(\delta)$ given in (4.3) except that they are multiplied by p, and that g and f are used in place of G and F. These latter substitutions simply reflect the fact that the econometrician is ignoring the periodic structure of the (conditional) autocovariances of $\{y_t\}$ when estimating the model.

Finally, consider $A_3(\delta)$. Substituting (4.8) into expression (4.3) for A_3 gives

$$A_3(\delta) = [\nu-\mu(\delta)]'G(0,\delta)^{-1}[\nu-\mu(\delta)]$$

$$= \sum_{j=0}^{p-1}[\nu-\mu(\delta)]'\alpha(2\pi j/p)g(2\pi j/p,\delta)^{-1}$$

$$\times\alpha(-2\pi j/p)'[\nu-\mu(\delta)]. \tag{4.11}$$

Not surprisingly, if neither ν nor $\mu(\delta)$ has a periodic structure (i.e., is constant across seasons), then the formula for $A_3(\delta)$ simplifies further to be p times the expression given in (4.3) with $G(0,\delta)$ replaced by $g(0,\delta)$.

Taken together, (4.9)–(4.11) give a representation for the approximation criterion when the model allows for a periodic structure only in the means but not in the autocovariances:

$$A(\delta) = \frac{p}{2\pi}\int_{-\pi}^{\pi}\left\{\log\det g(\omega,\delta) + \operatorname{trace}\left[g(\omega,\delta)^{-1}f(\omega)\right]\right\}\mathrm{d}\omega$$

$$+\sum_{j=0}^{p-1}[\nu-\mu(\delta)]'\alpha(2\pi j/p)g(2\pi j/p,\delta)^{-1}$$

$$\times\alpha(-2\pi j/p)'[\nu-\mu(\delta)]. \tag{4.12}$$

The approximation criterion (4.2) can be used to analyze the consequences of a variety of types of misspecification, of which we now describe three. First, let the true model be periodic, but assume that an econometrician ignores the periodic structure in the (conditional) autocovariances. Let the econometrician use the seasonally unadjusted data, and let him impose whatever restrictions his theory puts on the seasonal means of the $\{y_t\}$ process.

A second application is the same as the first, except that the economist first extracts seasonal means from the data by regressing on seasonal dummies. The researcher does not impose the restrictions that his theory imposes on the mean vector $\mu(\delta)$. In this situation, the relevant approximation criterion is simply $A_1(\delta)+A_2(\delta)$.

A third application is related to a setting that was described by Sims (1990). Let the true model have period $p=1$. Let the source of seasonality be exogenous seasonality, say, as reflected in either $a_b(L)$ or $a_d(L)$ in the example model of section 3. The seasonality is complicated in that the polynomial in L is quite long. The econometrician commits the misspecification of prematurely truncating this polynomial. This is his only specification error. Two possible procedures are to be compared. The first is to use seasonally unadjusted data, in which case approximation criterion (4.2) applies. The second procedure is to use seasonally adjusted data. For example, let the spectral density of the seasonally adjusted data be $F^a(\omega)=c(\mathrm{e}^{-i\omega})F(\omega)c(\mathrm{e}^{+i\omega})'$, where $c(L)$ is a diagonal matrix of (two-sided) seasonal adjustment filters. When the researcher uses seasonally adjusted data, $F^a(\omega)$ replaces $F(\omega)$ in approximation criterion (4.2).

In a similar context, Sims (1976, 1990) has argued that the second procedure can give superior estimates of the key economic parameters, e.g., the parameters of utility or technology in the examples of section 3. The argument is that the term $A_2(\delta)=(1/2\pi)\int_{-\pi}^{\pi}\operatorname{trace}[G(\omega,\delta)^{-1}F^a(\omega)]\,\mathrm{d}\omega$ assigns the most weight to approximation error at frequencies at which $F^a(\omega)$ [or $F(\omega)$, if unadjusted data are used] is large. Since $F(\omega)$ is large at precisely those frequencies where $G(\omega,\delta)$ is most misspecified, the effects of misspecification on the estimates of the parameters of preferences and technologies will be diminished by weighting by $F^a(\omega)$ rather than $F(\omega)$. This argument exploits the property that the economic model is a dynamic one that embodies the cross-equation and cross-frequency restrictions imposed by rational expectations.[11] Notice that it is *not* the intent of this procedure to improve

[11] Arrayed against Sims's (large sample) argument in favor of using seasonally adjusted data is a countervailing efficiency consideration. The strength of the overidentifying information imposed by the model is strongest at precisely the high-spectral-power frequencies that Sims proposes to suppress. This is because those restrictions are of the usual cross-frequency, cross-equation kind implied by rational expectations. Suppressing these frequencies via seasonal adjustment imposes an efficiency loss.

the estimates of the autoregressive polynomials $a_b(L)$ and $a_d(L)$. Instead, the argument is that by accepting *worse* estimates of these polynomials, one can improve estimates of other more interesting parameters.

This leads us to consider a notion of *ideal* seasonal adjustment. Suppose that we use a band-pass filter to eliminate the seasonal frequencies. Formally, this can be accomplished using a frequency-domain approximation to the likelihood function, as discussed by Hannan (1973) and Dunsmuir and Hannan (1976), except that some seasonal frequencies are omitted. In addition, suppose that the model is correctly specified so that there exists a parameter vector δ_0 such that $G(\cdot, \delta_0) = F$ at all frequencies except possibly for seasonal frequencies. Also assume that $\mu(\delta_0) = \nu$. Let C be an indicator function equal to one for all frequencies, except at the seasonal frequencies, and zero at the seasonal frequencies. The relevant approximation criterion is now given by

$$\begin{aligned} A(\delta) = \frac{1}{2\pi} \int_{-\pi}^{\pi} C(\omega) \Big\{ & \log\det G(\omega, \delta) \\ & + \operatorname{trace}\left[G(\omega, \delta)^{-1} F(\omega) \right] \Big\} \, \mathrm{d}\omega \\ & + [\nu - \mu(\delta)]' G(0, \delta)^{-1} [\nu - \mu(\delta)] . \end{aligned} \tag{4.13}$$

Using the familiar information inequality for normally distributed random variable [e.g., see Kullback and Leibler (1951)], we have that for each ω

$$\begin{aligned} & \log\det G(\omega, \delta) + \operatorname{trace}\left[G(\omega, \delta)^{-1} F(\omega) \right] \\ & \geq \log\det G(\omega, \delta_0) + \operatorname{trace}\left[G(\omega, \delta_0)^{-1} F(\omega) \right] \\ & = \log\det F(\omega) + \operatorname{trace}(I) . \end{aligned} \tag{4.14}$$

It follows that δ_0 is among the family of minimizers of A. Omitting frequencies might hinder identification by introducing additional minimizers. When the model remains identified, the omission of frequencies will typically increase the asymptotic variance of the resulting parameter estimator unless the seasonal frequencies are contaminated by measurement error.[12]

In the following section, we display several examples that serve to illustrate this argument. Other examples will illustrate other lessons as well.

[12]Strictly speaking, to avail ourselves of such an argument that consistency can be retained by ignoring seasonal frequencies, it is necessary to assume that the measurement error is *purely* seasonal. Such a measurement error process would be linearly deterministic, i.e., perfectly predictable linearly from its own past.

5. Examples

In this section, we apply versions of approximation criterion (4.2) to study the effects of using seasonally adjusted and unadjusted data to estimate several examples. Our first example focuses on the effects of using three types of data to estimate a model that is misspecified only because it ignores the periodic coefficient variation that is the true source of seasonality. Our second and third examples are designed to illustrate the effects of using seasonally adjusted data to estimate a correctly specified model, where the correct specification implies spectral peaks at seasonal frequencies. Two more examples illustrate the effects of estimating particular misspecified time-invariant models alternatively using seasonally adjusted and unadjusted data.

5.1. Estimating a time-invariant model when the data are generated by a periodic model

We assume that the truth is a version of a model that falls into category c of section 3, but that the econometrician fits a model of category a. In particular, the true model is (3.1)–(3.3) with the following parameter settings: $p=4$, $\gamma_1=0.13$, $\gamma_2=0.1$, $\gamma_3=0.1$, $\gamma_4=0.08$, $\phi_1=0.3$, $\delta_k=0.95$, $\lambda=0$, $\beta=1/1.05$, $a_b(L)=1-0.2L$, $a_d(L)=1-0.4L$, $\mathrm{E}w_{1t}^2=0.25^2$, $\mathrm{E}w_{2t}^2=1$, $\mathrm{E}b_t=30$, $\mathrm{E}d_t=5$.

The econometrician is assumed to fit a version of model (3.1)–(3.3) that is correct in all respects, except that it ignores the hidden periodic structure of the productivity parameter $\gamma_{s(t)}$. To capture seasonality in the data, the econometrician fits a fifth-order autoregressive process for the endowment process $\{d_t\}$. We further assume that the econometrician knows and imposes the true values of all but five of the parameters. The econometrician estimates the parameters γ, ϕ_1, a_1, a_4, and a_5, where the parameters a_j pertain to the following representation for the endowment process:

$$d_t=5+\tilde{d}_t,\qquad \tilde{d}_t=a_1\tilde{d}_{t-1}+a_4\tilde{d}_{t-4}+a_5\tilde{d}_{t-5}+w_{2t}.$$

Note that the econometrician imposes the correct value of the mean of the endowment process.

We compare estimates that would emerge from maximum likelihood applied to three different types of data: 1. seasonally unadjusted data; 2. data that are seasonally unadjusted, but from which deterministic means, estimated by regressing on seasonal dummies, have been removed; and 3. data that are 'ideally' seasonally adjusted. For the seasonally unadjusted data, approximation criterion (4.2) applies directly. For the data from which seasonal means have been extracted, (4.2) is modified by dropping the term $A_3(\delta)$, so that the appropriate approximation criterion becomes $A_1(\delta)+A_2(\delta)$. The case in which seasonally adjusted data are used is modelled by

Table 1
Estimated time-invariant model when the true model is periodic.

Parameter	Seasonally unadjusted	No means	Seasonally adjusted	Truth
γ	0.0996	0.1030	0.1025	$\gamma_{s(t)}$
ϕ_1	0.4132	0.2854	0.3007	0.3000
a_1	0.931	0.3250	0.3987	0.4000
a_4	0.4383	0.1598	−0.0003	0.000
a_5	−0.4298	−0.1261	−0.0005	0.000

using the approximation criterion $\tilde{A}_1(\delta)+\tilde{A}_2(\delta)$, where $\tilde{A}_j(\delta)$ is created by using a formula identical to that for $A_j(\delta)$, except that zero weight is assigned to frequencies at and surrounding the seasonal frequencies.[13] This description of $\tilde{A}_j(\delta)$ serves to define what we mean by 'ideally' seasonally adjusted data.[14]

We assume that the econometrician has data on $\{c_t, i_t\}$. We compute the estimates that the econometrician would make using each of the three types of data by numerically minimizing the appropriate version of (4.4) with N set at either 2^6 or 2^7. Results are reported in table 1 and figs. 1, 2, and 3. The results are best when the model is estimated using seasonally adjusted data, and worst when the seasonally adjusted data are used *without* extracting seasonal means. Table 1 shows that when seasonally unadjusted data are used, the estimated parameters falsely impute seasonality to the endowment process: note the values for a_1, a_4, a_5. This effect is worse when seasonal means are not removed. Note also how the adjustment cost parameter ϕ_1 is misestimated.

For the case in which seasonally unadjusted data are used but where seasonal means are first removed, fig. 2 reveals how the parameter estimates adjust to compromise among errors at different frequency bands. In order to diminish the errors at the seasonal bands, the estimates put some seasonality into $\{c_t, i_t\}$ the only way that the model permits, namely, through the endowment process. Doing that helps the model 'match' at the seasonal frequencies at the cost of worse fit at the nonseasonal frequencies.

Fig. 1 indicates that the match between approximate and true spectral densities is much worse with the seasonally unadjusted data when seasonal means are not extracted first. The action of the term $A_3(\delta)$ in (4.2) is the cause of this effect.

Fig. 3 and table 1 indicate that using seasonally adjusted data delivers excellent estimates of all parameter values. Note how well both ϕ_1 and the

[13]We applied zero weight to the five frequencies at and adjacent to $\pi/2$, and also the frequency π and the two next lowest frequencies.

[14]Thus, 'ideally' means application of an ideal band-pass filter. It is known that this is not ideal from the standpoint of a signal extraction model of seasonal adjustment; for example, see Grether and Nerlove (1970).

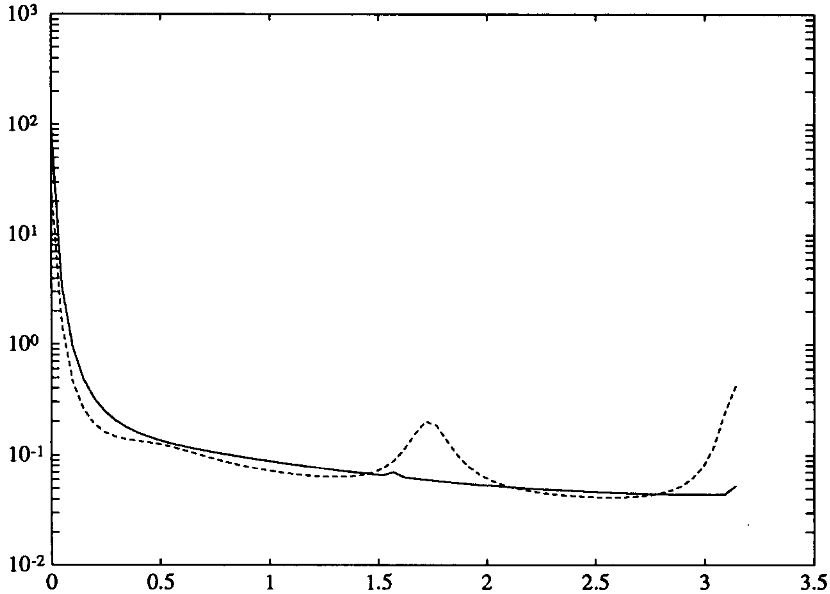

Fig. 1a. Spectrum of consumption for the true and the approximate (dotted line) models for the case in which seasonally unadjusted data are used and the true model is periodic.

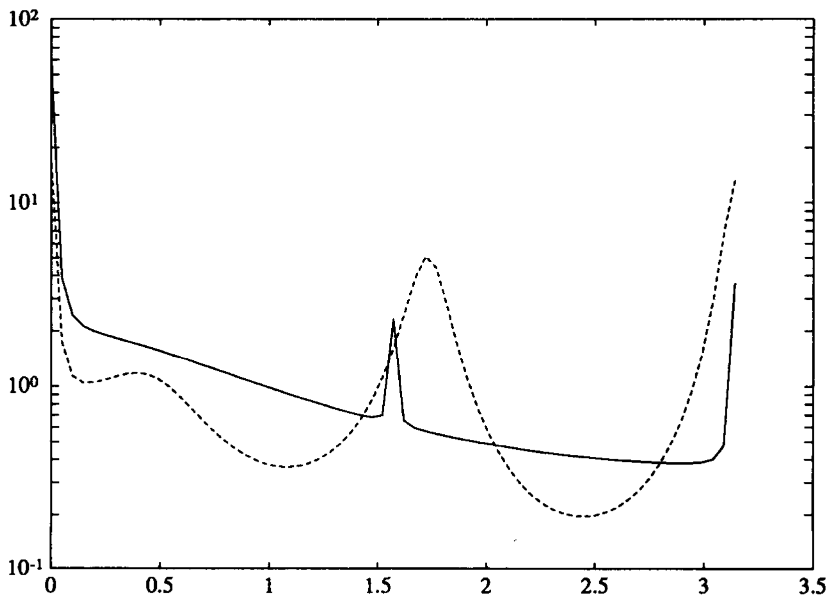

Fig. 1b. Spectrum of investment for the true and the approximate (dotted line) models for the case in which seasonally unadjusted data are used and the true model is periodic.

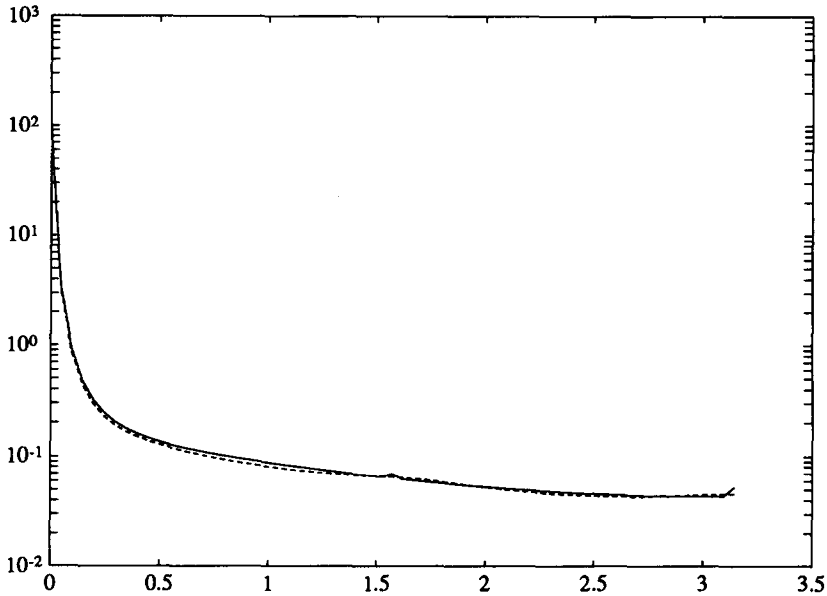

Fig. 2a. Spectrum of consumption for the true and the approximate (dotted line) models for the case in which seasonally unadjusted data with seasonal means removed are used and the true model is periodic.

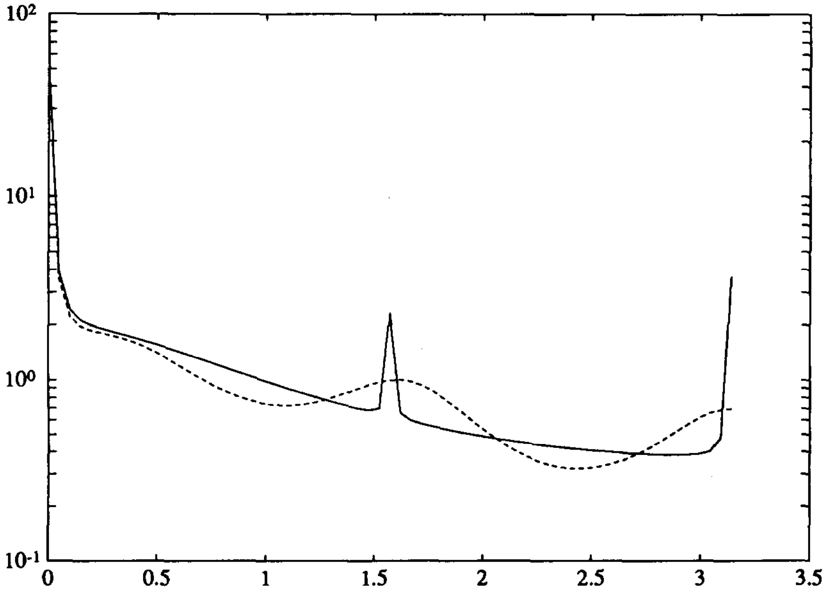

Fig. 2b. Spectrum of investment for the true and the approximate (dotted line) models for the case in which seasonally unadjusted data with seasonal means removed are used and the true model is periodic.

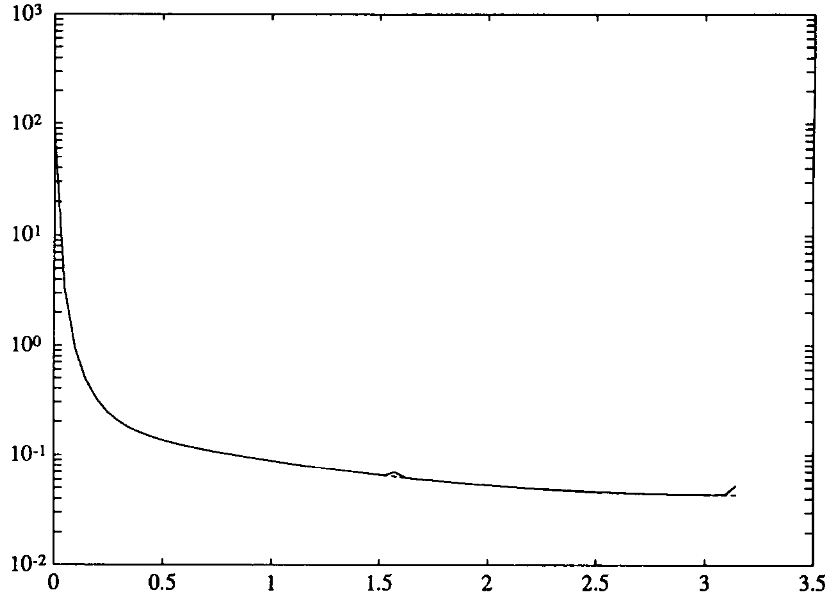

Fig. 3a. Spectrum of consumption for the true and the approximate (dotted line) models for the case in which seasonally adjusted data are used and the true model is periodic.

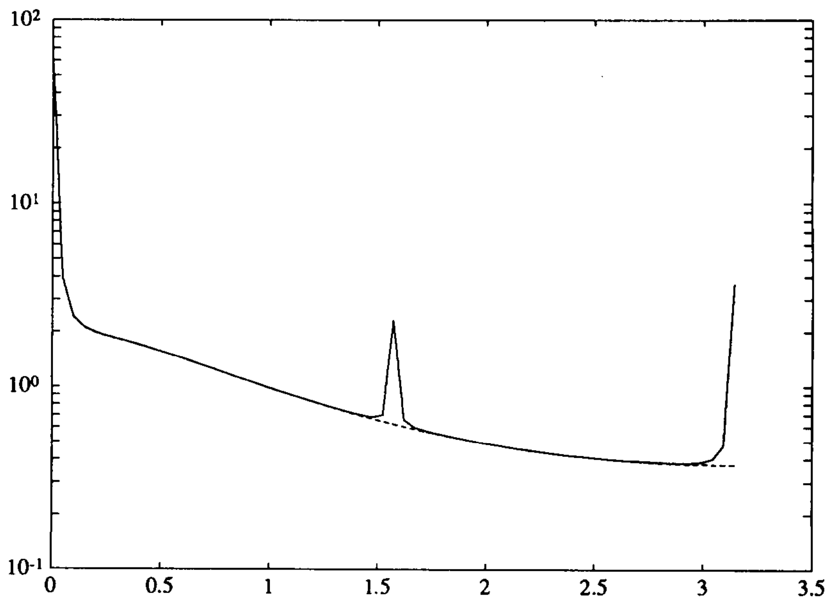

Fig. 3b. Spectrum of investment for the true and the approximate (dotted line) models for the case in which seasonally adjusted data are used and the true model is periodic.

stochastic process for $\{d_t\}$ are estimated. The parameter γ is estimated to be 0.1025, which is the arithmetic average of $\gamma_{s(t)}$. Fig. 3 indicates that the estimated model approximates the true model virtually perfectly at the nonseasonal frequencies.

5.2. *Estimating a time-invariant model with seasonal habit persistence using seasonally adjusted data*[15]

In this example, we assume that the truth is a time-invariant version of (3.1)–(3.3) with seasonal habit persistence. The parameters assume the values $\phi_1 = 0.005$, $\lambda = 0.8$, $\delta_h = 0.9$, $\gamma = 0.1$, $\delta_k = 0.95$, $\beta = 1/1.05$. We set the $\{d_t, b_t\}$ processes to satisfy

$$d_t = 5 + \tilde{d}_t, \qquad \tilde{d}_t = a_1 \tilde{d}_{t-1} + w_{2t},$$

$$b_t = 30 + \tilde{b}_t, \qquad \tilde{b}_t = 0.2\tilde{b}_{t-1} + w_{1t},$$

where $a_1 = 0.7$, w_{2t} is a white noise with unit variance, and w_{1t} is a white noise with variance 0.25^2. These settings for the parameters, especially for λ, δ_h, imply that consumption and investment will have spectral peaks at seasonal frequencies, as shown in fig. 4.

The econometrician is assumed to estimate the correct model on the basis of observations on $\{c_t, i_t\}$. He knows the true values of all parameters except $\delta_h, \lambda, \gamma, a_1, \phi_1$, each of which he estimates. For *each* of the three ways of estimating the model described with our preceding example, the econometrician recovers precisely the correct parameter values. Fig. 4 reports the true and approximating spectral densities for the case in which the econometrician uses seasonally adjusted data. Notice that they are equal (the dotted line lies atop the solid line).

Because the econometrician is fitting the correct model, it is evident that he will recover the true parameter values when he uses seasonally unadjusted data, whether or not seasonal means have been extracted. This is a consequence of the strong overidentification of the model delivered by the rational expectations restrictions. What might not be so immediately evident is that these overidentifying restrictions so strongly restrict the behavior of consumption and investment *across frequencies*, that even when the seasonal frequencies are dropped from the frequency-domain likelihood function, the approximation criterion is minimized by setting the parameters at their correct values. By setting the parameters at their correct values, $A(\delta)$ is

[15]In all of the following examples, we ignore the term $A_3(\delta)$ of the approximation criterion (4.2). This amounts to assuming that the econometrician does not impose the restrictions that his model implies about the means of the variables. Ignoring this term has no effect on the examples in subsections 5.2 and 5.3, but does affect those in subsections 5.4 and 5.5.

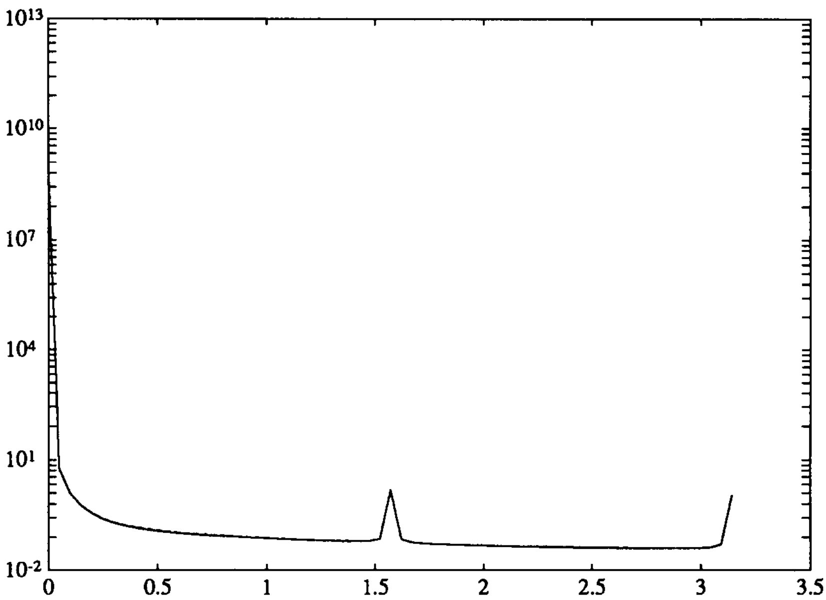

Fig. 4a. Spectrum of consumption for the true and the approximate (dotted line) models for the case in which seasonally adjusted data are used and the true model is time-invariant with seasonal habit persistence.

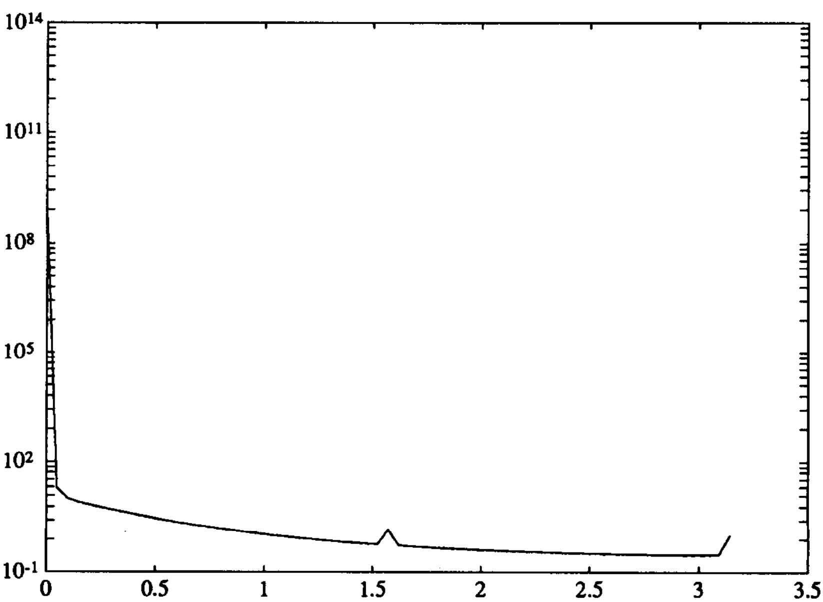

Fig. 4b. Spectrum of investment for the true and the approximate (dotted line) models for the case in which seasonally adjusted data are used and the true model is time-invariant with seasonal habit persistence.

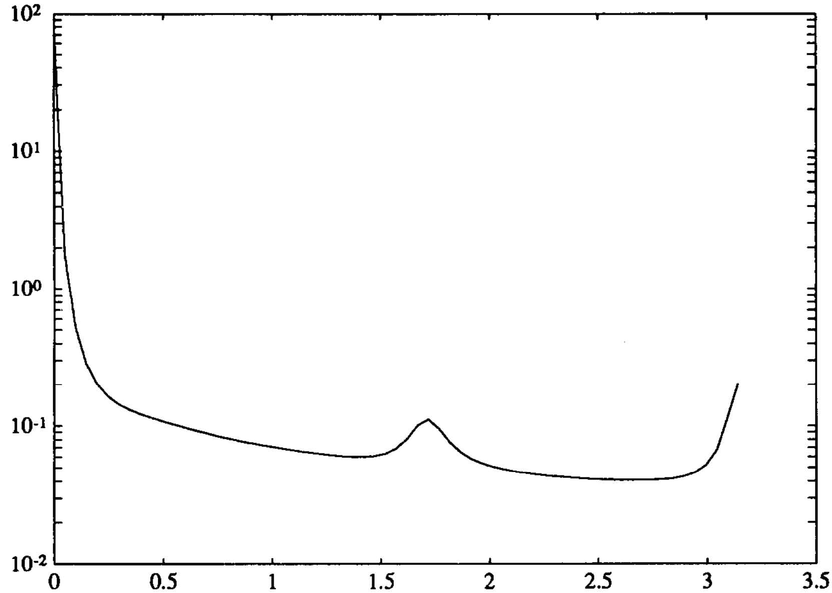

Fig. 5a. Spectrum of consumption for the true and the approximate (dotted line) models for the case in which seasonally adjusted data are used and the true model is time-invariant with seasonal endowment process.

minimized, even though the seasonal frequencies have been dropped, and even though the *main* role of the parameters λ, δ_h is to induce seasonal covariation between consumption and investment. While these parameters impinge most strongly on the seasonal frequencies, they affect *all* frequencies in ways that are sufficient to preserve their identification even when the seasonal frequencies are dropped.

5.3. *Estimating a time-invariant model with a seasonal endowment process, using seasonally adjusted data*

The true model is assumed to be a time-invariant version of (3.1)–(3.3), with the same parameter settings as used in section 5.1 except that now $\gamma_{s(t)} = 0.1$, $\forall t$, and d_t follows the process

$$d_t = 5 + \tilde{d}_t, \qquad \tilde{d}_t = a_1 \tilde{d}_{t-1} + a_4 \tilde{d}_{t-4} + a_5 \tilde{d}_{t-5} + w_{2t},$$

where w_{2t} is a white noise with unit variance and $a_1 = 0.1$, $a_4 = 0.5$, $a_5 = -0.4$. The econometrician estimates the correct model. The econometrician knows all the parameters except $\gamma, \phi_1, a_1, a_4, a_5$, each of which he estimates.

We study the consequences of giving the econometrician two alternative data sets. One contains observations on $\{c_t, i_t\}$, the other observations on

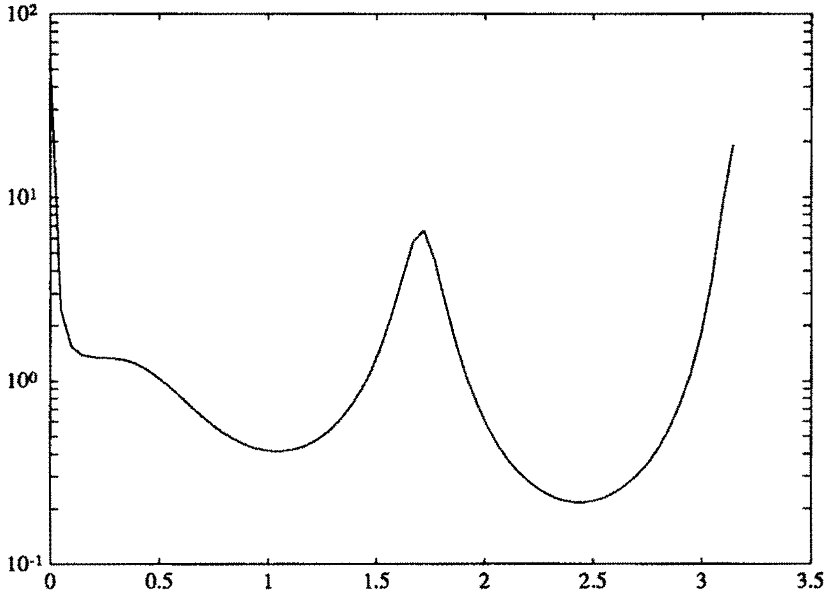

Fig. 5b. Spectrum of investment for the true and the approximate (dotted line) models for the case in which seasonally adjusted data are used and the true model is time-invariant with seasonal endowment process.

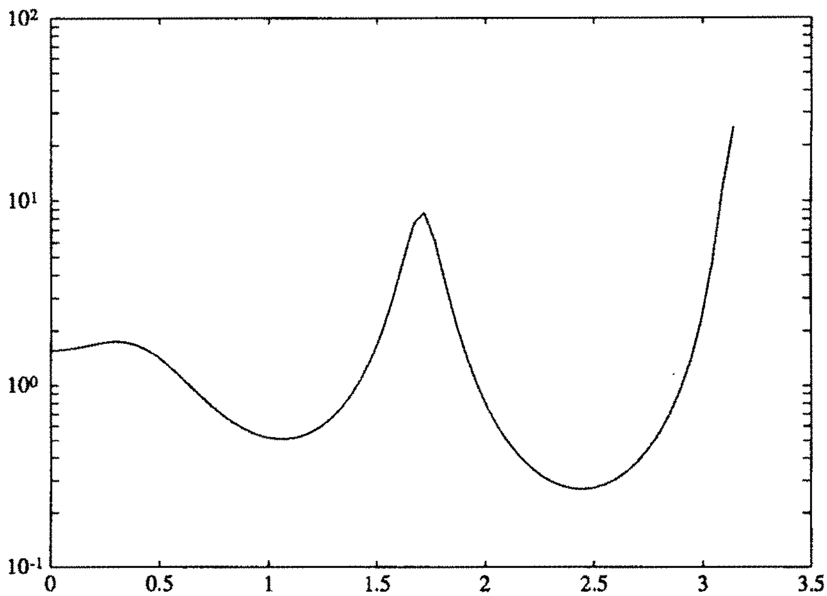

Fig. 5c. Spectrum of endowment process for the true and the approximate (dotted line) models for the case in which seasonally adjusted data are used and the true model is time-invariant with seasonal endowment process.

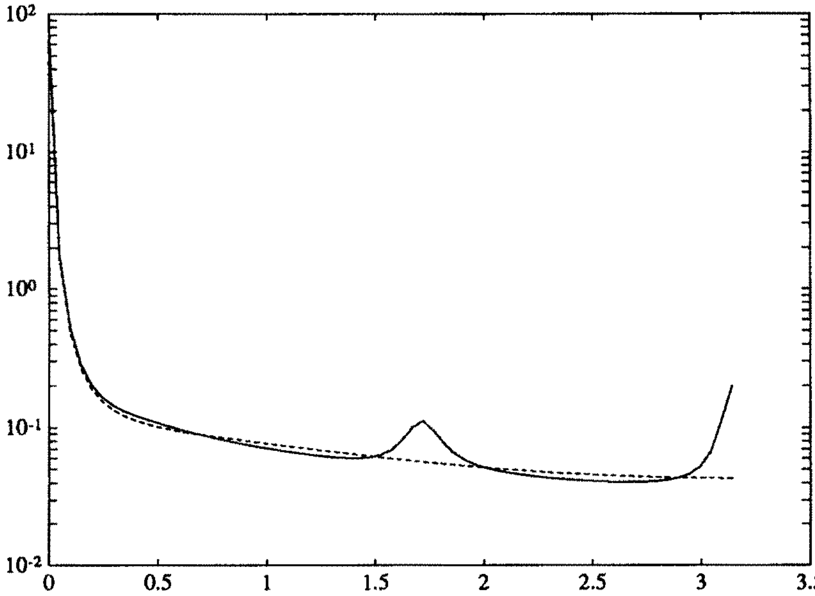

Fig. 6a. Spectrum of consumption for the true and the approximate (dotted line) models for the case in which seasonally adjusted data are used, the true model is time-invariant with seasonal endowment process, and the approximating model imposes a first-order autoregressive endowment shock.

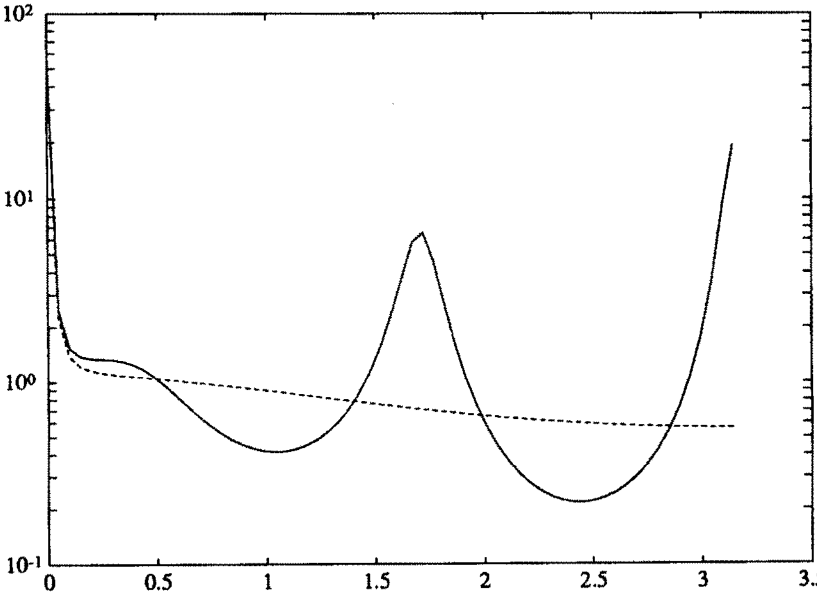

Fig. 6b. Spectrum of investment for the true and the approximate (dotted line) models for the case in which seasonally adjusted data are used, the true model is time-invariant with seasonal endowment process, and the approximating model imposes a first-order autoregressive endowment shock.

$\{c_t, d_t\}$.[16] In both cases, the econometrician correctly estimates all five parameters, regardless of which of our three procedures he employs. Fig. 5 shows the spectral densities of the observables for the true and approximating models when seasonally adjusted data are used. Figs. 5a and 5b illustrate the outcome for the data set $\{c_t, i_t\}$, while fig. 5c shows the spectral density for the true and approximating endowment process $\{d_t\}$. These results have the same explanation offered in section 5.2.

5.4. *Estimating a time-invariant model with too short an autoregressive process for the endowment process with seasonally adjusted and unadjusted data*

In this example, the true economy is identical with that used in section 5.3. Now the econometrician is assumed to be estimating a model that is misspecified only in the following respect: he imposes that $a_4 = a_5 = 0$, so that he estimates a first-order autoregressive process for the endowment shock. The econometrician is assumed to estimate the parameters γ, ϕ_1, a_1, and to know the values of the other parameters. The econometrician estimates the model using data on $\{c_t, i_t\}$. We calculated the econometrician's estimates on the assumption that he uses both seasonally unadjusted and 'ideally' seasonally adjusted data.

When the econometrician uses seasonally adjusted data, he recovers the estimates $\gamma = 0.1000$, $\phi_1 = 0.3011$, $a_1 = 0.1631$, the first two of which are close to their true values. Fig. 6 reports the spectral densities of consumption and investment for the true and best approximating models.

When the econometrician uses seasonally unadjusted data, he recovers the estimates $\gamma = 0.1003$, $\phi_1 = 0.2964$, $a_1 = -0.2018$. Although the estimates for the first two are close to their true values, they are worse than when seasonally adjusted data are used. Fig. 7 reports the spectral densities of consumption and investment for the true and the best approximating models. These figures indicate how the approximation criterion has seen fit to set a_1 to a negative value in order to try to match the peak in the spectral density at frequency π.

5.5. *Estimating a time-invariant model without seasonal habit persistence using seasonally adjusted data when the true model has seasonal habit persistence*

In this example, the true model is identical with that described in section 5.2. Now the econometrician fits a model that is misspecified because it erroneously sets $\lambda = 0$, $\delta_h = 0$. This is the only misspecification committed. The econometrician estimates the parameters γ, ϕ_1, a_1 using 'ideally' season-

[16]The case in which the econometrician directly observes the endowment shock matches up with a case studied in a variety of papers on linear rational expectations models, e.g., Hansen and Sargent (1980).

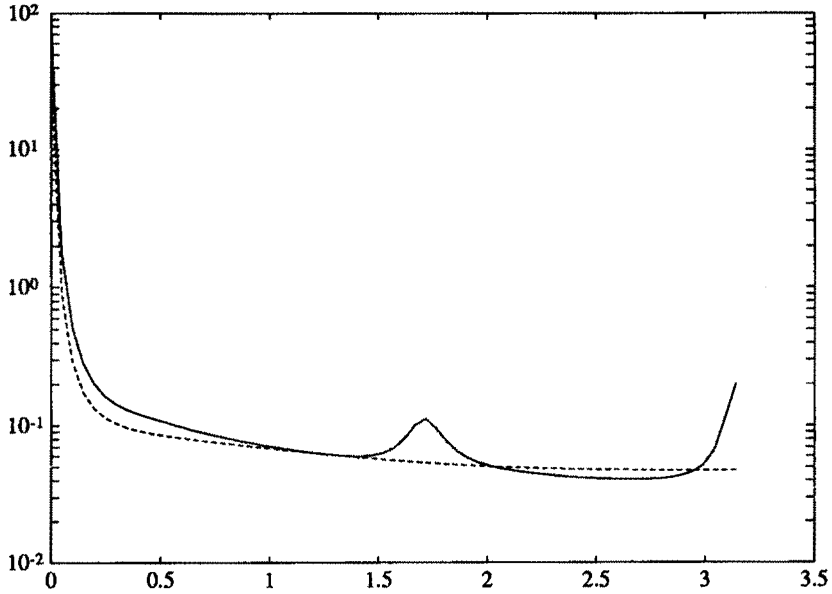

Fig. 7a. Spectrum of consumption for the true and the approximate (dotted line) models for the case in which seasonally unadjusted data are used, the true model is time-invariant with seasonal endowment process, and the approximating model imposes a first-order autoregressive endowment shock.

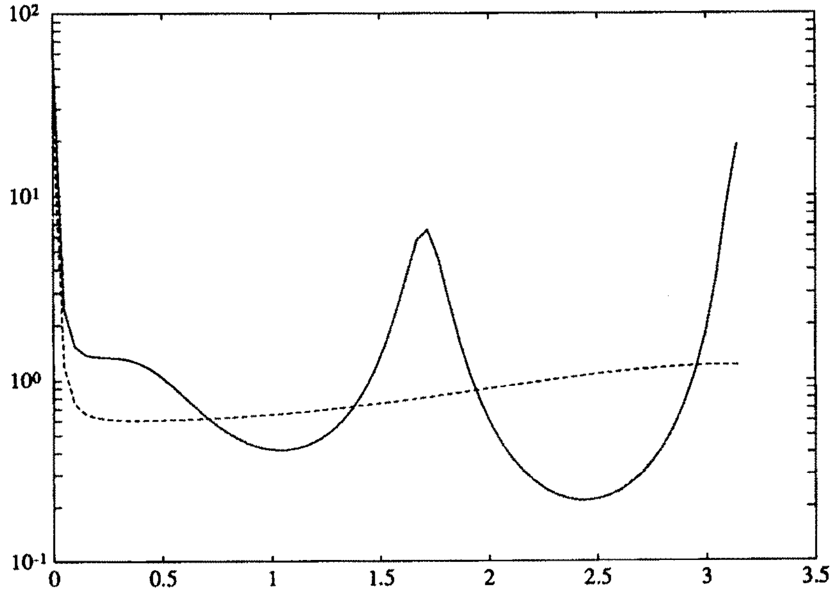

Fig. 7b. Spectrum of investment for the true and the approximate (dotted line) models for the case in which seasonally unadjusted data are used, the true model is time-invariant with seasonal endowment process, and the approximating model imposes a first-order autoregressive endowment shock.

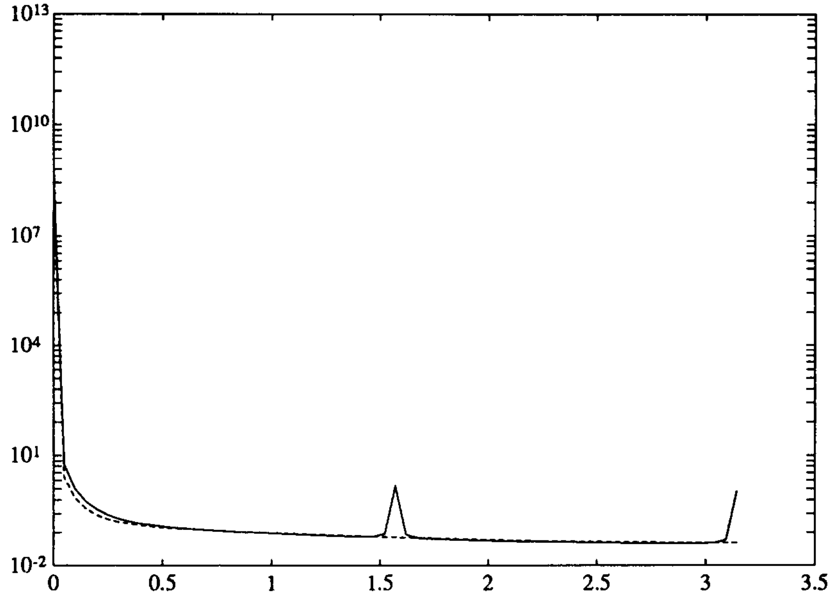

Fig. 8a. Spectrum of consumption for the true and the approximate (dotted line) models for the case in which seasonally adjusted data are used, the true model is time-invariant with seasonal habit persistence, and the approximating model ignores the seasonal habit persistence.

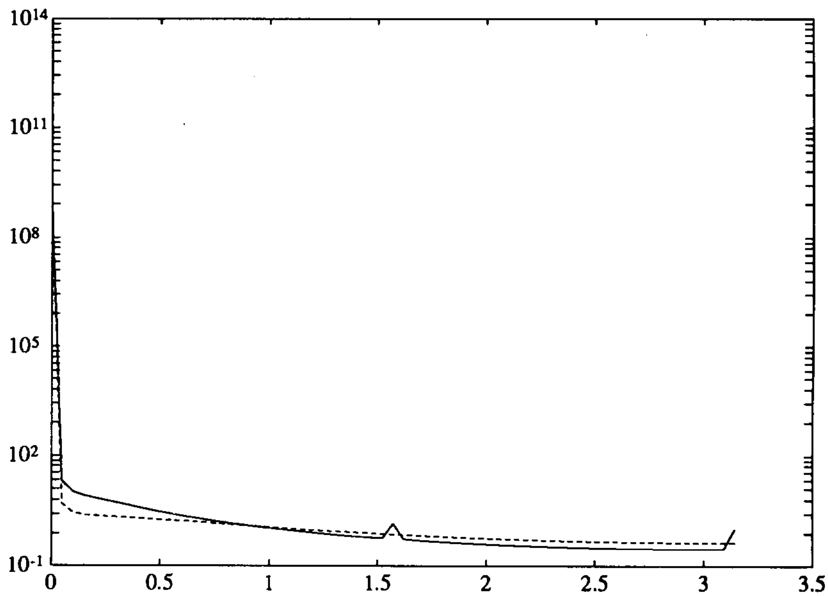

Fig. 8b. Spectrum of investment for the true and the approximate (dotted line) models for the case in which seasonally adjusted data are used, the true model is time-invariant with seasonal habit persistence, and the approximating model ignores the seasonal habit persistence.

ally adjusted data. We computed the econometrician's estimates to be $\gamma = 0.1000$, $\phi_1 = 0.0043$, $a_1 = 0.4019$. The 'technology parameters' γ, ϕ_1 are not far from their true values (0.1, 0.005), but the autoregressive parameter is far removed from its true value of 0.7. Fig. 8 depicts the spectral densities of consumption and investment of the true and the best approximating models.

6. Concluding remarks

Our examples illustrate the merit of Sims's recommendation that we estimate a rational expectations model by using seasonally adjusted data, especially when we are most unsure of the adequacy of the model's specification as it pertains to seasonal variation. The particular examples presented in this paper tend to suggest an asymmetry that favors Sims's recommendation: when the model is correctly specified, there is no loss of statistical consistency in estimating it with seasonally adjusted data; but when the model is misspecified, large reductions in asymptotic bias can emerge from using seasonally adjusted data.

This reading of our results should be tempered by the following considerations. First, the result that estimating a correctly specified model with seasonally adjusted data leads to no inconsistency has to be interpreted carefully. In particular, a correct specification has to be one that correctly captures both the nonseasonal and the seasonal variation, even though the seasonal frequencies will be ignored (or downplayed) during estimation. This interpretation is a consequence of the very cross-equation (and cross-frequency) restrictions which deliver the result of no inconsistency. It follows that this result will not necessarily rationalize modelling strategies that ignore explaining seasonal aspects of the data because seasonally adjusted data are to be used. Indeed, if the data are highly seasonal, one can appeal to this result only if the model being estimated itself implies that the data are highly seasonal in the proper way.

Second, when one is estimating a correctly specified model, there will occur a substantial efficiency loss from using seasonally adjusted data, a loss that grows with the importance of the seasonality in the raw data. In practice, there can be a trade-off between variance and bias of estimators, the dimensions of which we have not explored in this paper.[17]

Third, it is undoubtedly true that particular examples of estimation of misspecified models can go against Sims's recommendation.

Fourth, the examples computed in this paper all assume symmetric seasonal adjustment of all series. The issues of asymmetric seasonal adjustment studied by Sims (1974) and Wallis (1974) interact with the approximation issues studied here, and can be studied using our approximation criterion (4.2).

[17]On the basis of the logic leading to (4.2), something can be said about the asymptotic distributions of estimators of misspecified models. We plan to explore this in subsequent work.

While this paper has focused on using seasonally adjusted or unadjusted data to estimate approximate models, the approximation criterion (4.2) applies more broadly. Formally, the same arguments that support the use of seasonally adjusted data also apply to using data that have been filtered to eliminate, say, low-frequency components. For example, Hodrick and Prescott (1980) compose what is essentially a high-pass filter that eliminates very-low-frequency components from time series while leaving business cycle and higher frequencies intact. Prescott and many other researchers have tried to match data that have been filtered with the Hodrick–Prescott filter to theoretical models that they have designed explicitly to capture business cycle frequencies, but whose low-frequency properties they distrust. Such an empirical strategy can probably be rationalized through an appropriate application of our approximation criterion (4.2).

Appendix

In this appendix, we discuss more formally the approximation-error criterion introduced in section 4. A detailed treatment of this issue can be found in Pötscher (1987) where, among other things, the approximation results for correctly specified models in Hannan (1973), Dunsmuir and Hannan (1976), and Deistler, Dunsmuir, and Hannan (1978) are extended to accommodate model misspecification. Pötscher's analysis abstracts from the contribution of the mean term, $A_3(\delta)$, to the approximation error. Not surprisingly, it is straightforward to accommodate this additional contribution.

In truth, $\{Y_t\}$ is a process of the form described in section 2. It is not necessarily Gaussian. A researcher fits a possibly false approximating model using a Gaussian likelihood function parameterized by a vector δ in a parameter space Δ. For a sample of size N, let $X_N' \equiv [Y_1', Y_2', \ldots, Y_N']$ denote the data vector observed by the econometrician. Associated with the parameter δ is a mean vector $\mu(\delta)$ and a spectral density function $G(\cdot, \delta)$ for $\{Y_t\}$. The vector $\mu(\delta)$ implies a mean vector $\mu_N(\delta)$ for the composite random vector X_N constructed by simply stacking N copies of $\mu(\delta)$. The spectral density function $G(\cdot, \delta)$ implies a hypothetical covariance matrix $\Gamma_N(\delta)$ for $X_N - \mu_N(\delta)$. Apart from a constant term, the negative logarithm of the Gaussian likelihood function for the sample is given by

$$L_N(\delta) \equiv \tfrac{1}{2} \log \det \Gamma_N(\delta) + \tfrac{1}{2}[X_N - \mu_N(\delta)]' \Gamma_N(\delta)^{-1} [X_N - \mu_N(\delta)]. \tag{A.1}$$

To analyze formally the large sample properties of a misspecified likelihood estimator, we must say more about the parameterization of the mean and the spectral density function.

Assumption A.1. Δ is a compact subset of a Euclidean space.

Assumption A.2. G is a continuous function mapping $[-\pi, \pi] \times \Delta$ into the space of positive-definite Hermitian matrices such that for some $0 < \varepsilon_l < \varepsilon_\mu$, $\varepsilon_l I \leq G \leq \varepsilon_\mu I$, and for each $(\omega, \delta) \in [-\pi, \pi] \times \Delta$, $G(-\omega, \delta)$ is the complex conjugate of $G(\omega, \delta)$.

Assumption A.3. μ is a continuous function on the domain Δ.

In regards to Assumptions A.1–A.3, the existence of upper and lower bounds on G and $|\mu|$ over the entire parameter space is severe and is made for convenience. Pötscher (1987) described ways to relax Assumptions A.1 and A.2. It would also be of interest to relax Assumption A.3 to avoid requiring that the hypothetical mean vectors reside in a a compact set.

Consider the first term in the log-likelihood function. It follows from Dunsmuir and Hanna (1976, lemma 2) that $\{(1/N)\log\det \Gamma_N(\delta)\}$ converges uniformly to $A_1(\delta)$, where

$$A_1(\delta) \equiv \frac{1}{2\pi}\int_{-\pi}^{\pi} \log\det G(\omega, \delta)\,\mathrm{d}\omega. \tag{A.2}$$

To study the second term of the log-likelihood function, let ν_N be the vector obtained by stacking N identical copies of the mean ν of Y_t. Decompose the vector $[X_N - \mu_N(\delta)]$ into two components:

$$[X_N - \mu_N(\delta)] = [X_N - \nu_N] + [\nu_N - \mu_N(\delta)]. \tag{A.3}$$

We use this decomposition to form a corresponding decomposition of the second term on the right side of (A.1):

$$\begin{aligned}
&[X_N - \mu_N(\delta)]'\Gamma_N(\delta)^{-1}[X_N - \mu_N(\delta)] \\
&= [X_N - \nu_N]'\Gamma_N(\delta)^{-1}[X_N - \nu_N] \\
&\quad + 2[X_N - \nu_N]'\Gamma_N(\delta)^{-1}[\nu_N - \mu_N(\delta)] \\
&\quad + [\nu_N - \mu_N(\delta)]'\Gamma_N(\delta)^{-1}[\nu_N - \mu_N(\delta)].
\end{aligned} \tag{A.4}$$

The proof strategy adopted by Hannan (1973), Dunsmuir and Hannan (1976), and Deistler, Dunsmuir, and Hannan (1978), and Pötscher (1987) is to form a finite-order autoregressive approximation to the candidate model implied by $G(\cdot, \delta)$. More precisely, they form the Cesaro sum of a finite number of Fourier coefficients of $G(\cdot, \delta)^{-1}$. This results in an approximating function H^{-1} of G^{-1}, where the approximation can be made uniform over the compact set $[-\pi, \pi] \times \Delta$. For a sufficiently small choice of $\varepsilon > 0$, H^{-1}

can be chosen so that

$$0 < H^{-1} - \varepsilon I \leq G^{-1} \leq H^{-1} + \varepsilon I. \tag{A.5}$$

Let $\Phi_N(\delta)$ be the counterpart to $\Gamma_N(\delta)$ when spectral density $[H(\omega,\delta)^{-1} - \varepsilon I]^{-1}$ is used in place in $G(\omega,\delta)$. Similarly, let $\Psi_N(\delta)$ be the counterpart to $\Gamma_N(\delta)$ using $[H(\omega,\delta)^{-1} + \varepsilon I]^{-1}$ is used instead of $G(\cdot,\delta)$.

As noted by Dunsmuir and Hannan (1976, p. 351), for some positive number c,

$$0 \leq \Gamma_N(\delta)^{-1} - \Phi_N(\delta)^{-1} \leq (c\varepsilon) I_N,$$

$$0 \leq \Psi_N(\delta)^{-1} - \Gamma_N(\delta)^{-1} \leq (c\varepsilon) I_N, \tag{A.6}$$

where I_N is an identity matrix with the same dimensions as $\Gamma_N(\delta)$. It follows from the Law of Large Numbers for stationary ergodic processes and the boundedness of μ that for sufficiently large N and some positive number c^*,

$$0 \leq (1/N)[X_N - \mu_N(\delta)]'\left[\Gamma_N(\delta)^{-1} - \Phi_N(\delta)^{-1}\right]$$
$$\times [X_N - \mu_N(\delta)] \leq c^* \varepsilon,$$

$$0 \leq (1/N)[X_N - \mu_N(\delta)]'\left[\Psi_N(\delta)^{-1}\Gamma_N(\delta)^{-1}\right]$$
$$\times [X_N - \mu_N(\delta)] \leq c^* \varepsilon. \tag{A.7}$$

Inequalities (A.7) provide uniform bounds on the approximation error induced by using autoregressive approximations for the spectral density matrix $G(\cdot,\delta)$.

We now investigate the large sample behavior of the autoregressive approximations. We consider use of $[H(\omega,\delta)^{-1} + \varepsilon I]^{-1}$. For each δ, let $B(L;\delta)$ be the autoregressive polynomial of degree M used in the approximation so that

$$(H^{-1} + \varepsilon I)^{-1}(\omega,\delta) = B[\exp(-i\omega);\delta]B[\exp(i\omega);\delta]'. \tag{A.8}$$

Using arguments from Hannan (1973, lemmas 1, 4), Dunsmuir and Hannan (1976, lemma 3), and Pötscher (1987, lemma 3.7), it follows that $\{(1/N)\times$

$[X_N - \nu_N]'\Psi_N(\delta)^{-1}[X_N - \nu_N]$) converges almost surely to

$$A_{\varepsilon,2}(\delta) \equiv \frac{1}{2\pi}\int_{-\pi}^{\pi} \operatorname{trace}\left[H(\omega,\delta)^{-1} + \varepsilon I\right] F(\omega)\,\mathrm{d}\omega$$

$$\geq \frac{1}{2\pi}\int_{-\pi}^{\pi} \operatorname{trace} G(\omega,\delta)^{-1} F(\omega)\,\mathrm{d}\omega \equiv A_2(\delta), \tag{A.9}$$

uniformly in δ, where F is the true spectral density function. (Note that $A_{\varepsilon,2}$ depends on ε directly and through the construction of H^{-1}.)

Abstracting from initial conditions, $[X_N - \nu_N]'\Psi_N(\delta)^{-1}[\nu_N - \mu_N(\delta)]$ is given by

$$\sum_{t=M+1}^{N} \left[B(L;\delta)(Y_t - \nu)\right] \cdot \left\{B(1;\delta)\left[\nu - \mu(\delta)\right]\right\}. \tag{A.10}$$

Dividing the sum by N, it follows from the Law of Large Numbers for stationary, ergodic processes that this term converges to zero almost surely, uniformly in δ.

Again abstracting from initial conditions, the term $[\nu_N - \mu_N(\delta)]' \times \Psi_N(\delta)^{-1}[\nu_N - \mu_N(\delta)]$ for the autoregressive approximation can be represented as

$$\sum_{t=M+1}^{N} \left\{B(1;\delta)\left[\nu - \mu(\delta)\right]\right\} \cdot \left\{B(1;\delta)\left[\nu - \mu(\delta)\right]\right\}. \tag{A.11}$$

Dividing the sum in (A.11) by N, we see that this term converges uniformly to

$$A_{\varepsilon,3}(\delta) \equiv \left[\nu - \mu(\delta)\right]'\left[\left(H(0,\delta)^{-1} + \varepsilon I\right)\right]\left[\nu - \mu(\delta)\right]$$

$$\geq \left[\nu - \mu(\delta)\right]'G(0,\delta)^{-1}\left[\nu - \mu(\delta)\right]' \equiv A_3(\delta), \tag{A.12}$$

as N gets large.

A similar set of arguments applies for $[H(\omega,\delta)^{-1} - \varepsilon I]^{-1}$. In light of (A.7) and decomposition (A.4), for a sufficiently large N,

$$\begin{aligned}
&A_2(\delta) + A_3(\delta) + (c^* + 1)\varepsilon \\
&\geq A_{-\varepsilon,2}(\delta) + A_{-\varepsilon,3}(\delta) + (c^* + 1)\epsilon \\
&\geq (1/N)\left[X_N - \mu_N(\delta)\right]'\Gamma_N(\delta)^{-1}\left[X_N - \mu_N(\delta)\right] \\
&\geq A_{\varepsilon,2}(\delta) + A_{\varepsilon,3}(\delta) - (c^* + 1)\varepsilon \\
&\geq A_2(\delta) + A_3(\delta) - (c^* + 1)\varepsilon.
\end{aligned} \tag{A.13}$$

Since the original choice of ε is arbitrary, it follows that $\{(1/N)[X_N - \mu_N(\cdot)]'\Gamma_N(\cdot)^{-1}[X_N - \mu_N(\cdot)]\}$ converges uniformly to $A_2 + A_3$ almost surely. Combining these results we find that $\{(2/N)L_N\}$ converges uniformly to L_0 almost surely, where

$$L_0 \equiv A_1 + A_2 + A_3. \tag{A.14}$$

Let δ_N denote the set of minimizers of L_N and δ_0 the set of minimizers of L_0. Given the uniform convergence just established, with probability one

$$\limsup_{N\to\infty} \delta_N \subseteq \delta_0, \tag{A.15}$$

where our notion of convergence is that implied by the Hausdorff metric on the space of nonempty compact sets [e.g., see Hildenbrand (1974)]. When δ_0 is a singleton set, the lim sup can be replaced by lim.

References

Akaike, H., 1973, Information theory and an extension of the likelihood principle, in: B.N. Petrov and F. Csaki, eds., Proceedings of the second international symposium of information theory (Akademiai Kiado, Budapest).

Becker, Gary and Kevin Murphy, 1988, A model of rational addiction, Journal of Political Economy 96, 675–700.

Birchenhall, C.R., R.C. Bladen-Hovell, A.P.L. Chui, D.R. Osborn, and J.P. Smith, 1989, A seasonal model of consumption, Economic Journal 99, 837–843.

Breiman, Leo, 1968, Probability (Addison-Wesley, Reading, MA).

Brock, William A. and Leonard J. Mirman, 1972, Optimal economic growth and uncertainty: The discounted case, Journal of Economic Theory 4, 479–513.

Burridge, Peter and Kenneth F. Wallis, 1990, Seasonal adjustment and Kalman filtering: Extension to periodic variances, Journal of Forecasting 9, 109–118.

Cleveland, W.P. and G.C. Tiao, 1979, Modeling seasonal time series, Économie Appliquée XXXII, 107–129.

Cochrane, J.H., 1988, How big is the random walk in GNP?, Journal of Political Economy 96, 893–920.

Deistler, M., W. Dunsmuir, and E.J. Hannan, 1978, Vector linear time series models: Corrections and extensions, Advances in Applied Probability 8, 360–372.

Dunsmuir, W., 1978, A central limit theorem for parameter estimation in stationary linear time series and its application to models for signal observed with noise, Annals of Statistics 7, 490–506.

Dunsmuir, W. and E.J. Hannan, 1976, Vector linear time series models, Advances in Applied Probability 8, 339–364.

Gladysev, E.G., 1961, Periodically correlated random sequences, Sov. Math. 2, 385–388.

Grether, David and M. Nerlove, 1970, Some properties of 'optimal' seasonal adjustment, Econometrica 38, 682–703.

Hall, Robert E., 1978, Stochastic implications of the life cycle-permanent income hypothesis: Theory and evidence, Journal of Political Economy 86, 971–988.

Hannan, E.J., 1970, Multiple time series (Wiley, New York, NY).
Hannan, E.J., 1973, The asymptotic theory of linear time series models, Journal of Applied Probability 10, 130–145.
Hansen, Lars P., 1990, Calculating asset prices in three example economies, in: T.F. Bewley, ed., Advances in econometrics: Fifth world congress (Cambridge University Press, Cambridge) 207–243.
Hansen, Lars Peter and Thomas J. Sargent, 1980, Formulating and estimating dynamic linear rational expectations models, Journal of Economic Dynamics and Control 2, 7–46.
Hansen, Lars P. and Thomas J. Sargent, 1991, Two difficulties in interpreting vector autoregressions, in: Lars P. Hansen and T.J. Sargent, eds., Rational expectations econometrics (Westview Press, Boulder, CO) 77–119.
Hansen, Lars P. and Thomas J. Sargent, 1992, Recursive linear models of dynamic economies, Manuscript.
Heaton, John C., 1990, The interaction between time nonseparable preferences and time aggregation, Sloan working paper no. 3181-90-EFA (Massachusetts Institute of Technology, Cambridge, MA).
Hildenbrand, W., 1974, Core and equilibria of a large economy (Princeton University Press, Princeton, NJ).
Hodrick, R.J. and E.C. Prescott, 1980, Post-war U.S. business cycles: An empirical investigation, Revised working paper, Nov. (Carnegie-Mellon University, Pittsburgh, PA).
Jones, R.H. and W.M. Brelsford, 1967, Time series with periodic structure, Biometrika 54, 403–408.
Kullback, S. and R.A. Leibler, 1951, On information and sufficiency, Annals of Mathematical Statistics 22, 79–86.
Ljung, L., 1978, Convergence analysis of parametric identification methods, IEEE Transactions on Automatic Control 23, 770–783.
Ljung, L., 1987, System identification: Theory for the user (Prentice Hall, Englewood Cliffs, NJ).
Ljung, L. and P.E. Caines, 1978, Asymptotic normality of prediction error estimation for approximate system models, Stochastics 3, 29–46.
Lucas, Robert E., Jr. and Edward C. Prescott, 1971, Investment under uncertainty, Econometrica 39, 659–681.
Monin, A.S., 1963, Stationary and periodic time series in the general circulation of the atmosphere, in: M. Rosenblatt, ed., Proceedings of the symposium on time series analysis (Wiley, New York, NY) 144–151.
Osborn, Denise R., 1988, Seasonality and habit persistence in a life cycle model of consumption, Journal of Applied Econometrics 3, 255–266.
Osborn, Denise, 1990, The implications of periodically varying coefficients for seasonal time series processes, Journal of Econometrics, forthcoming.
Osborn, Denise R. and Jeremy P. Smith, 1989, The performance of periodic autoregressive models in forecasting seasonal U.K. consumption, Journal of Economic and Business Statistics 7, 117–128.
Osborn, Denise, A.P.L. Chui, J.P. Smith, and C.R. Birchenhall, 1988, Seasonality and the order of integration for consumption, Oxford Bulletin of Economics and Statistics 50, 361–377.
Pagano, M., 1978, On periodic and multiple autoregressions, Annals of Statistics 6, 1310–1317.
Pötscher, B.M., 1987, Convergence results for maximum likelihood type estimators in multivariate ARMA models, Journal of Multivariate Analysis 21, 29–52.
Ryder, H.E. and G.M. Heal, 1973, Optical growth with intertemporally dependent preferences, Review of Economic Studies 40, 1–31.
Sargent, Thomas J., 1978, Comments on 'Seasonal adjustment and multiple time series analysis' by K.F. Wallis, in: A. Zellner, ed., Seasonal analysis of economic time series (Bureau of the Census, Washington, DC) 361–363.
Sargent, Thomas J., 1987, Macroeconomic theory, 2nd ed. (Academic Press, New York, NY) 336–342.
Sims, C.A., 1972, Approximate prior restrictions in distributed lag estimation, Journal of the American Statistical Association 67, 169–175.
Sims, C.A., 1974, Seasonality in regression, Journal of the American Statistical Association 69, 618–626.

Sims, C.A., 1976, Personal letter to Thomas Sargent, Oct.
Sims, C. 1990, Rational expectations modeling with seasonally adjusted data, Manuscript, Dec.; see also Journal of Econometrics, this issue.
Singleton, Kenneth, J., 1988, Econometric issues in the analysis of equilibrium business cycle models, Journal of Monetary Economics 21, 361–386.
Stokey, Nancy L. and Robert E. Lucas, Jr., with Edward C. Prescott, 1989, Recursive methods in economic dynamics (Harvard University Press, Cambridge, MA).
Tiao, G.C. and M.R. Grupe, 1980, Hidden periodic autoregressive-moving average models in time series data, Biometrika 67, 365–373.
Todd, Richard, 1983, A dynamic equilibrium model of seasonal and cyclical fluctuations in the corn–soybean–hog sector, Unpublished Ph.D. thesis (University of Minnesota, Minneapolis, MN).
Todd, Richard, 1990, Periodic linear quadratic periodic models of seasonality, Journal of Economic Dynamics and Control 14, 763–796.
Wallis, Kenneth, 1974, Seasonal adjustment and relations between variables, Journal of the American Statistical Association 69, 18–32.
White, H., 1982, Maximum likelihood estimation of misspecified models, Econometrica 50, 1–26.